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A pre-trained multi-representation fusion network for molecular property prediction

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ABSTRACT

In the field of machine learning and cheminformatics, the prediction of molecular properties holds significant importance. Molecules can be represented in various formats, including 1D SMILES string, 2D graph, and 3D conformation. Numerous models have been proposed for different representations to accomplish molecular property prediction. However, most recent works have focused on one or two representations or combining embedding vectors from different perspectives in an unsophisticated manner. To address this issue, we present PremuNet, a novel **pre**-trained **mu**lti-representation fusion **net**work for molecular property prediction. PremuNet can extract comprehensive molecular information from multiple views and combine them interactively through pre-training and fine-tuning. The framework of PremuNet consists of two branches: a Transformer-GNN branch that extracts SMILES and graph information, and a Fusion Net branch that extracts topology and geometry information, called PremuNet-L and PremuNet-H respectively. We employ masked self-supervised methods to enable the model to learn information fusion and achieve enhanced performance in downstream tasks. The proposed model has been evaluated on eight molecular property prediction tasks, including five classification and three regression tasks, and attained state-of-the-art performance in most cases. Additionally, we conduct the ablation studies to demonstrate the effect of each view and the branch combination approaches.

1. Introduction

Molecular property prediction is critically important in the areas of bioinformatics [1] and cheminformatics [2]. Accurate feature representation of molecules enables the effective solution of essential tasks, such as drug discovery [3], drug-target interaction prediction [4], and drug–drug interaction prediction [5]. Molecules can be represented as graphs, with atoms as nodes and chemical bonds as edges. Consequently, Graph Neural Network (GNN) methods are widely employed for tasks like molecular embedding and property prediction.

In recent years, various methods have been proposed for predicting molecular properties, including MPNN [6], GCN [7], GAT [8], GIN [9], etc. These models are graph neural networks that extract molecular properties from 2D molecular graphs. Additionally, molecules can be represented as 1D symbolic sequences using the Simplified Molecular Input Line Entry System (SMILES) [10]. The advantage of using 1D sequence representation is the applicability of natural language processing models, such as SMILES-Transformer [11] and SMILES-BERT [12]. These models can be enhanced through pre-training on large-scale molecular datasets, improving their feature extraction capabilities for new molecules. Moreover, 3D structure is crucial for predicting molecular properties due to the presence of stereoisomers, which have identical graph connectivity but differ in spatial arrangements, such as optical isomers and cis-trans isomers. Stereoisomers exhibit distinct physical properties, pharmacokinetics, and bioactivity [13–16], which cannot be captured from 2D graphs. As a result, many researchers utilize 3D networks like SchNet [17], DimeNet [18], and PAiNN [19] to extract molecular features. These networks apply the 3D coordinates and characteristics of each atom to extract geometric information, producing outstanding predictions of quantum properties.

However, models that rely on a single dimension often fail to capture all molecular features. For example, the Message Passing Neural Network (MPNN) allows each atom to use only adjacent atom features

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for updating, making it difficult for the model to extract atomic interaction information for distant atoms within the graph [6]. While using self-attention layers like SMILES-BERT enables the model to calculate self-attention scores among all atoms, the model may potentially miss critical information due to the absence of 2D topological structures. Using 3D networks may be a relatively good solution, as they can extract features from both 2D topological structures and 3D spatial information simultaneously. However, obtaining precise 3D structures of molecules is a challenging and time-consuming task [20]. Although 3D networks are effective at extracting molecular features, they may lack versatility as most datasets only provide SMILES strings [21].

Several multi-view models have been proposed to address the challenges in molecular property prediction. FP-GNN [22] is designed to learn features from both molecular fingerprint and 2D molecular representation, resulting in superior performance compared to single GNNs. AdvProp [23] employs BERT, MPNN and SVM to extract complementary embeddings from molecules, which can be combined for downstream tasks. 3D Infomax [24], GraphMVP [25], and Unified [26] utilize pre-training with 2D molecular graphs and 3D coordinate information to learn 3D information from 2D structures. These models address the limitations of 2D networks in learning 3D information and the issue of downstream datasets lacking 3D coordinates. However, some models exhibit limited performance on 2D datasets (e.g., 3D Infomax), and others may lack information in certain dimensions, leading to suboptimal performance on specific datasets (e.g., GraphMVP does not perform well on the Clintox dataset). To date, existing models cannot effectively extract and integrate information from all three dimensions simultaneously.

In light of these challenges, we introduce PremuNet, a pre-trained multi-representation fusion network designed for molecular property prediction. PremuNet can extract molecular embeddings from molecular fingerprints, SMILES strings, 2D structures and 3D geometries simultaneously, to predict molecular properties. The PremuNet approach also addresses the issue of missing 3D coordinates in molecular property prediction tasks by pre-training on large-scale datasets containing both 2D and 3D information. The pre-training employs self-supervised learning techniques, enabling PremuNet to achieve better results on datasets that lack 3D coordinates.

Our model consists of two branches: the PremuNet for low-dimension (PremuNet-L) branch, which incorporates SMILES-Transformer and GNN, and the PremuNet for high-dimension (PremuNet-H) branch, responsible for extracting 2D-3D fusion representations. Both branches can benefit from pre-training to improve model performance. In PremuNet-L, we first extract molecular features and atomic features from SMILES strings, and then combine atomic features with graphic information as the input of GNN, in order to obtain the 1D-2D fusion representations. Finally, we combine the molecular features, fusion representations, and molecular fingerprints to form the output of this branch. The PremuNet-H branch employs self-supervised pre-training techniques to facilitate the model's ability to learn relationships between the 2D structure and 3D coordinates of a molecule. Furthermore, the model employs 2D structures to produce 3D features, allowing it to acquire 3D features even in the absence of 3D coordinate data. In downstream tasks, features from both PremuNet-L and PremuNet-H branches are combined for prediction. After evaluating multiple methods, concatenating low-dimension and high-dimension features was experimentally proven to be the most effective approach.

To evaluate the effectiveness of our model in predicting molecular properties, we test PremuNet on eight datasets comprising five classification tasks and three regression tasks. Experimental results indicate that our approach achieves state-of-the-art performance in seven out of eight molecular property prediction tasks, with an average improvement of 3.4% for classification tasks and 4.0% for regression tasks compared to the previous best method. This demonstrates the effectiveness of our multi-representation pre-training network. Furthermore, we conduct comprehensive ablation experiments to elucidate the contribution of each module within our model and examine the impact of module fusion on the outcomes. Our codes are available at: https://github.com/A-Gentle-Cat/PremuNet.

Our contributions can be summarized as follows:

- Our paper presents a novel and advanced molecular multi-modal framework that predicts molecular properties utilizing two complementary molecular representations with distinct pre-trained methods.
- We introduce a novel multi-modal molecular fusion interaction scheme that employs GNN to effectively integrate molecular characteristics of various modalities.
- Extensive experiments on eight molecular prediction tasks are conducted to evaluate the performance of PremuNet. Compared with several state-of-the-art methods, PremuNet achieves significantly better performance in seven out of eight tasks.

2. Related work

2.1. Single-modal molecular representation learning

Previous works have developed various modalities to depict molecules in computer science, including SMILES strings, molecular graphs and 3D geometry [27]. The SMILES representation is compatible with Natural Language Processing (NLP) techniques as it represents the atoms, bonds and rings that compose a molecule as a string sequence. Numerous RNN- [28–30] and Transformer-related models [11,12] have been proposed for SMILES-based molecular representation learning. However, SMILES representation lacks the capability to express spatial chemical structure information, while graph-based methods are superior at modeling complex internal connectivity from molecular graphs [31]. Generally, most existing GNN models used for molecular graph learning follow a MPNN framework [6]. The MPNN framework consists of two steps: a message passing phase and a readout phase. To mine more structural information from molecular graphs, several MPNN variants were proposed, which achieved promising performance. CD-MVGNN [32] employs MPNN on atom- and bond-oriented graph structures for multi-view molecular representation learning. FraGAT [33] presents a fragment-based multi-scale graph attention model to extract features at three hierarchical levels of molecule structures. The molecule is actually a dynamic 3D structure beyond a simple 2D molecular graph, and contains complex geometric elements such as atomic positions, distances, and bond torsion angles [34]. In recent years, research on incorporating geometric information into graph architectures to benefit molecular property estimation tasks has attracted attentions [35,36]. For example, GEM [36] designed a geometry-based graph neural network architecture and several dedicated geometry-level self-supervised learning strategies to gain molecular geometry knowledge. Besides, Mol-BERT [37] proposed a novel pre-training framework with masked atoms modeling, enhancing the effectiveness of the pre-trained GNN. Unfortunately, since none of the above modalities can effectively construct a comprehensive molecular portrait, multi-modal molecular representation learning is paid more attention for its better performance than single-modal methods.

2.2. Multi-modal molecular representation learning

Inspired by advancements in the multi-modal learning field, the fusion of molecular features from different representation perspectives has become a popular direction in molecular property prediction. Recent studies mainly focus on designing sophisticated multi-modal learning frameworks [24–26] and modal interaction strategies [22,23,38]. For example, both GraphMVP [25] and 3D Infomax [24] utilize contrastive learning as a pretraining framework to maximize the mutual information between the embeddings of different molecular modal representations. Zhu et al. [26] propose the unified 2D and 3D pre-training framework that jointly leverages the 2D graph structure of the

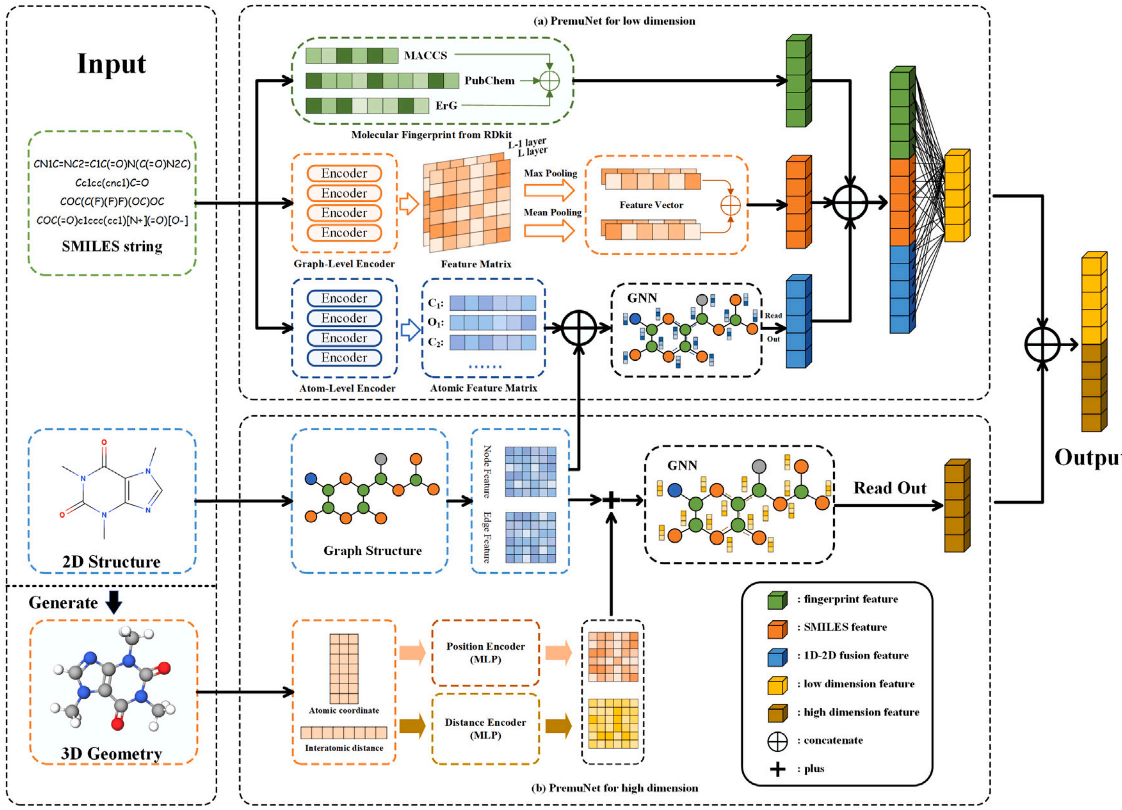

Fig. 1. An overview of PremuNet. The input comprises information from three modalities of the molecule: the SMILES string, 2D structure, and 3D geometry. (a) PremuNet-L comprises three components: Molecular Fingerprints, SMILES-Transformer, and GNN. Each component generates feature vectors, which are subsequently combined to represent the low dimensional (both 1D and 2D) features of molecules. (b) PremuNet-H module is designed to extract high-dimensional molecular features (both 2D and 3D) by integrating 2D and 3D information through the application of GNN.

molecule and the 3D conformation with mask strategy. MMSG [38] cross-references chemical information between molecular graphs and corresponding SMILES sequences to boost the representational power of molecular semantic features. Furthermore, several recent studies focus on the multimodality between natural language texts and molecules graphs [39], and GIT-Mol [40] proposes a multi-modal large language model that can process graphs, images, and texts simultaneously.

3. Method

3.1. PremuNet

PremuNet comprises two branches: PremuNet-L and PremuNet-H, designed to extract low-dimensional and high-dimensional molecular features respectively, as depicted in Fig. 1. In the PremuNet-L branch, we apply three methods to extract features from SMILES strings, including molecular fingerprints, SMILES-Transformer and GNN. We use the encoder of SMILES-Transformer to acquire atomic-level and molecular-level feature embeddings, and then use a GNN to implement 1D-2D information fusion. In the PremuNet-H branch, we employ a pre-trained unified GNN to input both 2D structures and 3D geometries simultaneously. Using this approach, we can achieve the fusion of 2D and 3D information. After that, the features derived from both branches are integrated to predict molecular properties.

3.2. PremuNet-L

In the PremuNet-L branch, we employ molecular fingerprints, SMILES-Transformer and GNN to extract low-dimensional features. For the molecular fingerprints, we use three different fingerprints with complementary information, which is delineated in Section 3.2.1. For the SMILES-Transformer, we calculate two feature matrices from tokenized SMILES strings, from which we can extract SMILES feature vectors and atomic feature vectors. After that, we combine atomic features and 2D topological information as the input of GNN, to achieve 1D-2D fusion molecular feature vectors. Finally, the molecular fingerprints, SMILES feature vectors and molecular feature vectors are concatenated to form the output of the PremuNet-L.

3.2.1. Molecular fingerprints

We utilize three distinct types of chemical molecular fingerprints: MACCS [41], PubChem [42], and ErG [43]. These fingerprints effectively extract the comprehensive characteristics of molecules, providing complementary information [22]. The detailed description of each fingerprint is as follows.

MACCS The MACCS fingerprint algorithm characterizes molecular structures by extracting their structural and functional features. This includes various chemical properties such as rings, chains, functional groups, lone pairs of electrons. These features are subsequently represented as a 166-bit binary code. By integrating the MACCS fingerprint with classifiers, the activity of diverse chemical compounds can be efficiently evaluated.

PubChem PubChem fingerprints are generated utilizing the PubChem dataset. These fingerprints involve matching a series of substructure fragments with molecules, and subsequently setting the corresponding binary positions to 1 or 0 based on the presence or absence of these substructures. This process results in the formation of an 881-bit binary code. PubChem fingerprints encompass a vast array of chemical information, enabling the swift comparison and screening of compounds within compound libraries. Consequently, they play a crucial role in drug development, drug screening, and chemical information retrieval.

ErG. The ErG molecular fingerprint identifies critical aspects of molecular and receptor interactions, including hydrogen bond properties, hydrophobic regions of specific dimensions, and noteworthy aromatic and aliphatic rings. ErG allocates these features to attribute points with distinct characteristics, transforms them into radial distribution functions, and ultimately integrates them into a descriptor vector. This molecular fingerprinting technique is employed in areas such as small molecule virtual screening.

3.2.2. SMILES-Transformer

In this study, we utilize the SMILES-Transformer structure to extract molecular and atomic feature vectors from SMILES strings. We extract feature vectors from SMILES strings using the following steps: SMILES tokenization, self-attention information calculation and feature vector extraction. The SMILES strings must first be tokenized before they can be input into the Transformer structure. Pertinent information can be extracted from the SMILES string.

1. Atomic information, such as "O", "N", "Br", etc.
2. Charge information, such as "+2", "−3", etc.
3. Chemical bond information, such as "=" (double bond), "#" (triple bond)
4. Chirality information, such as "@@" and "@"
5. Ring information (represented by paired numbers)

We have developed Algorithm 1 to tokenize a SMILES string. Firstly, we use RDKit to convert the SMILES string to a 2D molecule, and then get the atom list $\mathcal{L}$ from the 2D molecule. It should be noted that the order of atoms in $\mathcal{L}$ is the same as that in the SMILES string. Employing an atom list helps to eradicate ambiguity in SMILES strings. For example, "Sc" can be understood as both "S" and "C" atoms, or as a single "Sc" atom. We check if the prefix of the SMILES string aligns with a token and repeat this process until the entire SMILES string is tokenized. Once this tokenization is complete, we obtain the list of tokens. In Algorithm 1, "$\pm Number$" represents charge information, such as "+2". $\mathcal{L}_i$ means the ith atom in atom list $\mathcal{L}$. $Prefix(\mathcal{S})$ means all prefixes of $\mathcal{S}$. "$Single\ Number$" means ring information, such as "11" (Although it appears in pairs in the SMILES string, we can only add one number to the token list at a time). $\mathcal{S}_0$ means the first element of $\mathcal{S}$. In addition, $\mathcal{S}-$ "@@" means remove the prefix "@@" from $\mathcal{S}$. The usage of other "−" symbols carries the same implication.

After the extraction process, we append "<BEG>" to the beginning and "<END>" to the end of the token list. Additionally, we pad the list with "<PAD>" until it reaches a predetermined length, denoted as N. To input the token list into the model, we use a dictionary to convert the token list into the sequence representation $v \in \mathbb{R}^N$. The converting dictionary can be seen at Appendix A. Then we use the Embedding Layer to convert the sequence into a feature matrix $V \in \mathbb{R}^{N\times hidden}$, where each $V_i \in \mathbb{R}^{hidden}$ represents the feature vector of v_i and $hidden$ is the length of hidden embedding.

Besides, because Transformer completely abandons the sequential structure of Recurrent Neural Network (RNN), we need to add the Position Embedding P to X to distinguish tokens in different positions. Formula (1), (2) can be used to calculate position embedding vector:

$$P_{(pos,2i)} = \sin\left(\frac{pos}{10000^{2i/hidden}}\right) \tag{1}$$

$$P_{(pos,2i+1)} = \cos\left(\frac{pos}{10000^{2i/hidden}}\right) \tag{2}$$

where pos represents the position, and i represents the dimension. The final Position Embedding matrix $P \in \mathbb{R}^{N\times hidden}$. In the end, we add the feature matrix and the position embedding matrix together to obtain the model input $X^{(0)} \in \mathbb{R}^{N\times hidden}$.

$$X^{(0)} = V + P \tag{3}$$

Afterwards, the matrix X is put into multiple layers of Transformer blocks for feature extraction. Each Transformer block consists of two parts, a self-attention layer and a feedforward neural network. The self-attention layer allows each position to learn contextual information, while the feedforward neural network can further extract feature information. We apply scaled dot-product attention to calculate attention score. Formally, the self-attention result matrix is calculated using the formula (4), (5) and the feed forward neural network is calculated with formula (6).

Algorithm 1 SMILES Tokenization

Input: SMILES string $\mathcal{S}$ of length N;
  Atom List $\mathcal{L}$ of size M;
  function $Prefix(\cdot)$
Output: Token List $\mathcal{T}$

$i \leftarrow 0$
$\mathcal{T} \leftarrow \{\}$
while $\mathcal{S}$ is not empty **do**
  if $i \neq M$ and $\mathcal{L}_i \in Prefix(\mathcal{S})$ **then**
    Append $\mathcal{L}_i$ to $\mathcal{T}$
    $i \leftarrow i + 1$
    $\mathcal{S} \leftarrow \mathcal{S} - \mathcal{L}_i$
  else if "$\pm Number$" $\in Prefix(\mathcal{S})$ **then**
    Append "$\pm Number$" to $\mathcal{T}$
    $\mathcal{S} \leftarrow \mathcal{S}-$"$\pm Number$"
  else if "@@"$\in Prefix(\mathcal{S})$ **then**
    Append "@@" to $\mathcal{T}$
    $\mathcal{S} \leftarrow \mathcal{S}-$"@@"
  else if "$Single\ Number$"$\in Prefix(\mathcal{S})$ **then**
    Append "$Single\ Number$" to $\mathcal{T}$
    $\mathcal{S} \leftarrow \mathcal{S}-$"$Single\ Number$"
  else
    Append $\mathcal{S}_0$ to $\mathcal{T}$ {Match a single symbol}
    $\mathcal{S} \leftarrow \mathcal{S} - \mathcal{S}_0$
  end if
end while
return $\mathcal{T}$

$$Attention(Q,K,V) = Softmax\left(\frac{Q \cdot K^T}{\sqrt{d_k}}\right)V \tag{4}$$

$$Softmax(z_i) = \frac{e^{z_i}}{\sum_{c=1}^{C} e^{z_c}} \tag{5}$$

$$FNN(x) = max(0, xW_1 + b_1)W_2 + b_2 \tag{6}$$

where $Q = XW^Q$, $K = XW^K$, $V = XW^V$, and W^Q, W^K, W^V, W_1, W_2, b_1, b_2 are trainable parameters, d_k is the dimension of Q and K. Moreover, in order to improve the performance, we use multi-head attention to accomplish self-attention calculation. Multi-head attention can be calculated using formula (7), (8).

$$MultiHead(Q,K,V) = Concat(head_1, \dots, head_h)W^O \tag{7}$$

$$head_i = Attention(QW_i^Q, KW_i^K, VW_i^V) \tag{8}$$

where h is the number of attention head and $W_i^Q, W_i^K, W_i^V \in \mathbb{R}^{hidden\times \frac{hidden}{h}}$, $W^O \in \mathbb{R}^{hidden\times hidden}$ are new parameter matrices.

After n layers of Attention Encoder feature extraction, we obtained the feature matrix of the SMILES string, $X^{(n)} \in \mathbb{R}^{N\times hidden}$. Here, $X_i^{(n)}$ represents the feature vector corresponding to the ith token. We apply multi-head attention layers, feed forward neural networks and normalization layers to update $X^{(i)}$, which can be seen in Fig. 2.

After obtaining the feature matrices $X^{(1)} \dots X^{(n)}$, we can get molecular features and atomic features from them. Molecular features consist of four parts, each with the same length. They are: $X_0^{(n)}$, $X_0^{(n-1)}$, the average value of all $X_i^{(n)}$ for each i and the maximum value of all $X_i^{(n)}$ for each i. We choose these feature embeddings because Lee

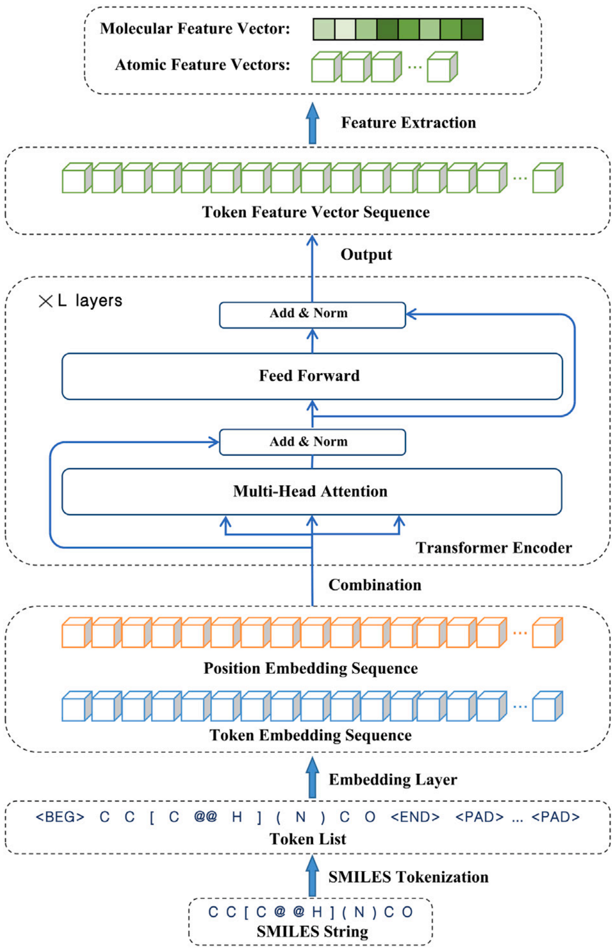

Fig. 2. The SMILES-Transformer process utilizes multi-layer Transformer encoders to extract molecular and atomic features from SMILES strings.

et al. [44] empirically showed the benefits of combining mean and max pooling. After that, we get the molecular feature vector of the SMILES string. Besides, we extract the feature vectors of all atoms from the corresponding indexes of $X^{(n)}$, which will be provided to the GNN as part of the input.

The process of extracting molecular and atomic features using SMILES-Transformer is depicted in Fig. 2.

3.2.3. Pretraining method of SMILES-Transformer

To improve the feature extraction ability, two pre-training tasks are developed for SMILES-Transformer. One task employs the AutoEncoder method, utilizing pre-training weights and dictionaries proposed by Honda et al. [11] to tokenize the original SMILES string. The encoding process involves passing the encoded string through multi-layer encoders, resulting in a feature matrix. Then, multi-layer decoders are used to decode the matrix to obtain the output, and the difference between input and output is minimized. This pre-training method allows the transformer encoder to effectively extract features from the SMILES string.

Another pretraining task involves unsupervised masked learning, which aims to extract atomic features from a SMILES string. For this part, we randomly mask some elements in the token list, inputting into the same encoder and decoder to obtain the output sequence, and minimize the difference between the output sequence and the token list that has not been masked. This pre-training method allows multi-layer encoders to learn the connections between tokens and be used to obtain atomic features.

Table 1
Chemical characteristics of atoms and bonds generated by RDKit.

Type	Feature	Size	Description
Node	Atomic num	119	Atomic type
	Chirality list	4	Chiral isomer.
	Degree list	12	Degree of nodes
	Charge list	12	Charges on atoms
	numH list	10	Adjacent hydrogen atoms
	Radical e list	6	Radical electrons
	Hybridization list	7	Hybridization orbital type
Edge	Is aromatic list	2	Aromatic bond or not
	Is in ring	2	In ring or not

For both of the pre-training methods mentioned above, we use cross-entropy loss function to calculate the value of loss. For each position, we use formula (9) for calculation, where C represents the number of token types, y represents the value from the original token list, and $\hat{y}$ represents the output value of the model.

$$\mathcal{L} = -\sum_{i=0}^{C} y_i \log \hat{y}_i \tag{9}$$

3.2.4. Graph neural network

Our GNN serves two primary functions: extracting graph features of molecules and fusing atomic feature vectors generated by the SMILES-Transformer. We utilize PNA (Principal Neighbourhood Aggregation) [45] as our GNN framework, and it can be replaced if necessary. PNA is a GNN that uses multiple aggregators and distinguishes neighbors with different features but different cardinalities through scalers. The PNA is defined as follows, using the MPNN framework.

$$X_i^{t+1} = U(X_i^{(t)}, \bigoplus_{(j,i)\in E} M(X_i^{(t)}, e_{j,i}, X_j^{(t)})) \tag{10}$$

$$\oplus = \begin{pmatrix} I \\ S(D, a=1) \\ S(D, a=-1) \end{pmatrix} \otimes \begin{pmatrix} \mu \\ \sigma \\ max \\ min \end{pmatrix} \tag{11}$$

$$S(d, \alpha) = (\frac{log(d+1)}{\delta})^{\alpha} \tag{12}$$

where $e_{j,i}$ denotes edge features, M and U are both neural networks (e.g., MLP), $\oplus$ denotes the aggregation operation, $\otimes$ denotes the tensor product, and the $\oplus$ operation is the tensor product of two matrices. The initial X is the combination of the chemical feature of the atom and the atomic feature provided by the SMILES-Transformer. In formula (11), the first matrix represents a scaler, and the second matrix represents an aggregation method. For the aggregation method, μ denotes the average aggregation, σ denotes the standard deviation aggregation, and max and min respectively denote maximum and minimum aggregations. Scaler S is used to multiply with the aggregation value during the aggregation operation, to amplify or reduce the message, in order to distinguish neighborhood points with the identical features but different cardinalities. After several iterations of the aforementioned message passing, the initial atomic features X will be updated to the 1D and 2D integrated atomic features.

In our model, we integrate information from two different modalities through PNA. The first one includes the chemical characteristics of atoms and bonds, generated by RDKit, as detailed in Table 1. The second one comprises atomic features produced using the SMILES-Transformer, as introduced in Section 3.2.2. We concatenate the atomic features generated by RDKit and the atomic feature vectors generated by SMILES-Transformer as the initial node features, and use the chemical bond features generated by RDKit as the initial edge features. Then, we input these node features, edge features, and topological structure into PNA network. Finally, we aggregate the vectors generated by the PNA network into molecular feature vectors with a readout method.

3.3. PremuNet-H

PremuNet-H is a unified framework based on GNN for extracting 2D and 3D information [26,46,47]. It can fuse 2D graph information and 3D geometry information to extract high-dimensional molecular features. The network consists of two parts: encoder and decoder. The encoder serves the function of integrating 2D and 3D information to achieve enhanced molecular feature representations. The decoder is employed for both pre-training tasks and downstream tasks.

In this paper, we used three pre-training tasks [26]:

- Mask all the atoms and reconstruct the masked atoms using 3D coordinates.
- Mask all the 3D coordinates and reconstruct the coordinates using atoms.
- Independently mask the atoms and the atomic coordinates with a probability of p and reconstruct the masked atoms and coordinates using the features of unmasked atoms and 3D coordinates.

The first and second tasks respectively equip the model with the ability to extract 3D information from 2D data and to extract 2D information from 3D data. The third task integrates these abilities, enhancing the model's capacity for information fusion. Through these three pre-training tasks, PremuNet-H can fully explore the high-dimensional features of molecules and encode them into feature vectors.

In the following, we will introduce the details of three pre-training tasks separately.

3.3.1. Atom reconstruction based on conformation

We mask all chemical characteristics of atoms (Details of the atomic characteristics are provided in Table 1), and only use the 3D coordinates of all atoms as input to predict all atomic characteristics information. Note that the topological structure of the atomic graph is preserved, but chemical characteristics of atoms and edges is masked. For each atomic characteristics, we regard it as a classification task and use the cross-entropy function as the loss function, which is defined as follows:

$$\mathcal{L}_{3D\to 2D} = -\sum_{i=1}^{m} y_i \log \hat{y}_i \tag{13}$$

where m is the number of categories for the prediction task. We independently use the above loss function to calculate the loss of each characteristics of the atom and then sum them up.

Following the aforementioned pre-training process, the model acquires the capability to extract 2D graphical information.

3.3.2. Conformation reconstruction based on atomic features

The objective of this pre-training task is to equip the model with the capability to generate 3D coordinates for molecules and extract relevant 3D information. We mask all atomic 3D coordinates and only use the original atomic features as input to predict and generate the 3D conformation of molecules. The loss function is defined as follows:

$$\mathcal{L}_{2D\to 3D} = \sum_{i\in V} \|R_i - \varrho(\alpha(\hat{R}_i))\|^2 \tag{14}$$

where $R_i \in \mathbb{R}^{1\times 3}$ represents the coordinate of the ith atom, and $\hat{R}_i \in \mathbb{R}^{1\times 3}$ represents the 3D conformation generated by the model. The ϱ operation is defined as $\varrho = RQ + t$, where $Q \in \mathbb{R}^{3\times 3}$ represents the rotation matrix, and $t \in \mathbb{R}^{1\times 3}$ represents the translation vector. Through this operation, $\mathcal{L}_{2D\to 3D}$ possesses rotational and permutation invariance.

3.3.3. Masked atom and conformation reconstruction

For the input molecule graph G, the set of all atoms in the molecule graph is denoted as V. For all $v_i \in V$, a fixed probability of p is used to perform masking. The same process is applied to the conformation. In fact, the mask task is a soft combination of the first two pre-training tasks, where the first two pre-training tasks represent the extreme cases of the mask task. It is worth noting that when calculating the loss function, only the masked part is used to compute the difference between prediction and ground truth, while the information that is not masked is not included in the loss function. The loss function of the mask task is defined as follows:

$$\mathcal{L}_{mask} = -\sum_{i\in \mathcal{I}_a}^{m} y_i \log \hat{y}_i + \sum_{i\in \mathcal{I}_b} \|R_i - \varrho(\alpha(\hat{R}_i))\|^2 \tag{15}$$

The definition of the symbols of formula (15) is covered in formula (14) and formula (13) and will not be repeated here.

3.3.4. PremuNet-H architecture

Encoder Our model, like most other GNN models, can be described using the MPNN framework [6]. For nodes, we use $x_i \in \mathbb{R}^d$ to represent the node information of the ith atom, where d represents the dimension. χ_i represents the extended feature including 3D information of the ith atom. Firstly, we encode the coordinates of the atom through a Multi-Layer Perceptron (MLP) to obtain a node-level 3D embedding information, and add it to the atomic feature to obtain the extended atomic feature χ_i.

$$\chi_i^{(l)} = x_i^{(l-1)} + MLP(R_i^{(l-1)}) \tag{16}$$

The features of the edge are represented by e_{ij}, where i and j are two connected nodes. Similar to the extraction of the 3D coordinates of the nodes, we encode the Euclidean distance between the 3D coordinates of nodes i and j using MLP, which are connected by the edge $e_{i,j}$. This encoder transforms the distance into an edge feature that contains 3D distance information. Next, we add the edge feature to the distance encoding information to obtain an extended edge feature ϵ_{ij}, which has 3D information.

$$\epsilon_i^{(l)} = e_{ij}^{(l-1)} + MLP(\|R_i^{(l-1)} - R_j^{(l-1)}\|) \tag{17}$$

Later, we will introduce how to update nodes and edges features. The m message in MPNN framework is defined as follows:

$$m_i = MLP(\chi_j^{(l-1)}; e_{ij}^{(l)}) \tag{18}$$

The method for updating nodes is as follows:

$$x_i^{(l)} = x_i^{(l-1)} + W_{node}(x_i^{(l-1)}; \widetilde{x}_i; u^{l-1}) \tag{19}$$

$$\widetilde{x}_i^{(l)} = \sum_{j\in \mathcal{N}_i} \alpha_j m_i \tag{20}$$

where α_j represents the attention score, and $\widetilde{x}_i$ represents the result of aggregating neighboring nodes using the attention mechanism. The attention score a_j is calculated as follows:

$$\alpha_j = Softmax_{i\in \mathcal{N}_j}\left(W_{atn}\xi(W_q\chi_i^{(l)} + W_k[\chi_j^{(l)}; \epsilon_{ij}^{(l)}]) \right) \tag{21}$$

The encoding process of atomic features is completed after L layers of the aforementioned message passing process, where W_q, W_k, W_{atn} are all learnable parameters.

The encoder loads pre-trained weights to enable the model to extract 2D and 3D information. Through a unified GNN framework, the encoder module encodes the input 2D and 3D information into a high-dimension feature fusion vector for downstream tasks.

Table 2
The details of the downstream task dataset.

Dataset	Instance	Description	3D %	Evaluation
BBBP	2039	Prediction of the barrier permeability of molecules.	94.0	ROC-AUC
BACE	1513	Prediction of binding results for a set of inhibitors of human β-secretase 1	99.9	ROC-AUC
Clintox	1477	Prediction of clinical trial toxicity and FDA approval status.	97.6	ROC-AUC
Tox21	7831	Prediction of qualitative toxicity measurements on 12 biological targets.	94.9	ROC-AUC
SIDER	1427	Prediction of marketed adverse drug reactions.	83.3	ROC-AUC
ESOL	1128	Prediction of water solubility data for common organic small molecules.	99.7	RMSE
Freesolv	642	Prediction of hydration free energy of small molecules in water.	99.8	RMSE
Lipophilicity	4200	Prediction of experimental results of octanol distribution coefficient.	99.6	RMSE

Table 2 provides detailed information regarding the downstream task datasets utilized in our study. The specifics include the size of each dataset, a description of the tasks involved, the proportion of 3D information extracted from the GEOM dataset, and the metric employed for evaluation.

Decoder. The decoder of the model consists of two parts: $AttributeDecoder$ and $PosDecoder$. They are used to decode the encoded information into the training targets. During the pre-training phase, $AttributeDecoder$ utilizes stacked MLPs to decode the atomic embeddings into the atomic attributes required for atomic reconstruction task. Similarly, $PosDecoder$ decodes the atomic embeddings into atomic coordinates for use in the conformation reconstruction task. During the fine-tuning phase, $AttributeDecoder$ drops the pre-training weights and uses random initialization to fine-tune on new downstream tasks.

4. Experiments

In the experimental section, we will first introduce the dataset used for pre-training and evaluation. Then, we introduce the parameter settings of the model during the pre-training process. Finally, we conduct a comparative experiment of our model against other baselines and analyze the experimental results.

4.1. Dataset

For the SMILES-Transformer that extracts molecular features, we pre-train on the ChEMBL database [48], which is a chemical database of bioactive molecules with drug-like properties, including 1.63 million unlabeled SMILES data. For the PremuNet-H, we use PCQM4Mv2 [49] as pre-training dataset for the PremuNet-H network. PCQM4Mv2 is a quantum chemistry dataset consisting of 3.38 million molecules, providing SMILES information, 2D graph information, and 3D geometric information. For the split of dataset, we use 95% of the molecules for training and 5% for validation.

For downstream tasks, we evaluate our model on five classification tasks and three regression tasks provided by MoleculeNet [21]. The classification tasks include: BBBP, BACE, Clintox, Tox21 and SIDER; while the regression tasks include: ESOL, FreeSolv, and Lipophilicity. Datasets of these tasks only contain 2D information. Accordingly, we utilized the method proposed by Axelrod and Gomez Bombarelli [50] to generate precise 3D molecular conformations from existing 1D SMILES data. A significant portion of the 3D conformational information of the molecules is obtained from the database provided by them. The specific details of datasets are provided in Table 2. For molecules without 3D information, we initialize their 3D coordinates with a uniform random distribution. All datasets are divided into 8:1:1 using the scaffold split provided by OGB [49]. For each evaluation of all models, we use the fixed split. All datasets and splits are available Appendix B.

4.2. Settings

4.2.1. PremuNet-L pretrain configuration

The two SMILES-Transformers are pre-trained with the same structure but different datasets and training strategies. During pre-training, a 4-layer Transformer structure is utilized, including encoder and decoder layers. Each layer has a hidden size of 256 and 4 attention heads for information aggregation.

For the Transformer that extracts molecular features, we use the Adam optimizer for self-supervised learning with a learning rate of 10^{-4}. We use cross entropy loss to compute the difference between input and output sequences. Since no masking is applied to the original sequence during the training, the value of loss decreases to 0 after 12 epochs, meaning perfect decoding from the encoded representations.

For the Transformer that extracts atomic features, we pre-train it on PCQM4Mv2 dataset using masked self-supervised learning. We mask 15% tokens of the token list as input and use the same hyper-parameters and model architecture. After training for several epochs, the model is capable of extracting the features of each token effectively.

4.2.2. PremuNet-H pretrain configuration

In the pre-training phase, the three pre-training tasks are run independently, and the results of the loss function are summed and back-propagated together. The number of GNN layers is set to 12, the vector dimension of the hidden layer is set to 256, the initial learning rate is set to 2×10^{-4}, the batch size is 128, the mask rate p is set to 0.25.

4.2.3. Finetune configuration

During the finetuning stage of the model, PremuNet-H and SMILES-Transformer respectively load pre-trained weights, while other parts are initialized randomly. It is worth noting that in PremuNet-H, the decoder for pre-training tasks is discarded and randomly initialized. We utilize concatenation for aggregation, and a detailed discussion of different aggregation methods is presented in Section 5. It is important to note that the GNN module is interchangeable. In our evaluation, PNA serves as the GNN. For the parameter settings, we adjust and test the model's batch size, learning rate, number of GNN layers, and hidden layer vector dimension. The detailed parameters are provided in our code.

4.3. Evaluation

We utilize ROC-AUC (Area under the ROC Curve) as the evaluation metric for classification tasks, and RMSE (Root-Mean-Square Error) for regression tasks. For multi-task datasets, we average the evaluation results of all the tasks. Each model is evaluated independently five times with random seeds. The mean and standard deviation of the results are recorded.

4.4. Baseline

We conduct a comprehensive evaluation of various popular models using different modalities to fully demonstrate the superiority of our model performance. The baselines we select are divided into two categories: uni-modal models and multi-modal models. Uni-modal models include those that only use SMILES sequence information, 2D graph information, or molecular 3D conformation information. Multi-modal models refer to models that simultaneously use multi-modal information.

Table 3
Model evaluation results on MoleculeNet. We highlight the best results in **bold** and the second best results in underline.

Method	BBBP	BACE	TOX21	Clintox	SIDER	ESOL	Freesolv	Lipophilicity	Avg_{cla}	Avg_{reg}
PNA	$65.0_{\pm1.0}$	$76.3_{\pm2.3}$	$61.2_{\pm0.4}$	$80.8_{\pm2.5}$	$57.6_{\pm0.5}$	$0.963_{\pm0.020}$	$2.894_{\pm0.895}$	$0.815_{\pm0.020}$	68.2	1.557
SMILES-TSFM[a]	$63.7_{\pm0.5}$	$76.0_{\pm0.9}$	$66.6_{\pm0.3}$	$98.2_{\pm0.2}$	$58.7_{\pm0.5}$	$1.339_{\pm0.017}$	$3.097_{\pm0.032}$	$1.215_{\pm0.016}$	72.6	1.884
CD-MVGNN	$69.3_{\pm0.7}$	$83.9_{\pm0.6}$	**$77.0_{\pm0.2}$**	$88.8_{\pm6.2}$	$61.1_{\pm1.0}$	$0.865_{\pm0.012}$	$2.048_{\pm0.217}$	$0.669_{\pm0.014}$	76.0	1.194
GROVER	$67.9_{\pm0.2}$	$79.1_{\pm3.4}$	$73.5_{\pm0.3}$	$73.1_{\pm6.8}$	$61.2_{\pm1.3}$	$0.992_{\pm0.096}$	$2.470_{\pm0.337}$	$0.810_{\pm0.033}$	71.0	1.424
Mol-BERT	$69.0_{\pm0.6}$	$79.7_{\pm1.4}$	$76.2_{\pm0.3}$	$79.4_{\pm1.3}$	$62.6_{\pm0.7}$	$1.211_{\pm0.047}$	$2.878_{\pm0.081}$	$0.732_{\pm0.008}$	73.4	1.607
GraphMVP	$69.4_{\pm0.9}$	$72.9_{\pm3.4}$	$75.4_{\pm0.3}$	$72.6_{\pm8.9}$	$61.8_{\pm0.9}$	$1.341_{\pm0.004}$	$2.972_{\pm0.137}$	$0.769_{\pm0.005}$	70.4	1.694
AdvProp	$72.2_{\pm0.6}$	$82.2_{\pm1.7}$	$75.2_{\pm5.0}$	$78.6_{\pm2.2}$	$60.3_{\pm7.8}$	$0.886_{\pm0.016}$	$1.996_{\pm0.251}$	$0.715_{\pm0.010}$	73.7	1.199
3DInfomax	$67.7_{\pm1.9}$	$80.7_{\pm2.9}$	$65.3_{\pm1.7}$	$88.5_{\pm2.0}$	$60.9_{\pm1.6}$	$0.901_{\pm0.025}$	$2.601_{\pm0.533}$	$1.226_{\pm0.970}$	72.6	1.576
MEMO[b]	$71.6_{\pm0.1}$	$82.6_{\pm0.3}$	$76.7_{\pm0.4}$	$81.6_{\pm3.7}$	$61.2_{\pm0.6}$	$0.984_{\pm0.034}$	/	$0.707_{\pm0.001}$	74.7	/
PremuNet-L	$72.5_{\pm0.2}$	$83.9_{\pm0.3}$	$72.2_{\pm0.3}$	$95.0_{\pm1.7}$	**$64.8_{\pm2.8}$**	$1.034_{\pm0.020}$	$1.905_{\pm0.018}$	$0.764_{\pm0.006}$	77.7	1.234
PremuNet-H	$65.2_{\pm0.9}$	$81.4_{\pm1.6}$	$73.9_{\pm0.6}$	$73.3_{\pm1.4}$	$58.8_{\pm1.4}$	$0.871_{\pm0.018}$	$2.265_{\pm0.078}$	**$0.629_{\pm0.008}$**	70.5	1.255
PremuNet	**$73.3_{\pm0.6}$**	**$84.3_{\pm0.5}$**	$74.0_{\pm1.2}$	**$99.2_{\pm0.2}$**	$62.6_{\pm0.8}$	**$0.851_{\pm0.038}$**	**$1.858_{\pm0.205}$**	$0.730_{\pm0.010}$	**78.7**	**1.146**

[a] SMILES-TSFM refers to the SMILES-Transformer model proposed by Honda et al [11].

[b] The authors of the MEMO do not make their code available. Since they utilized the same dataset and split we applied in our experiments, we directly used their data as presented in their paper. Notably, their data did not incorporate Freesolv, resulting in an unpopulated section in our analysis.

[c] Avg_{cla} and Avg_{reg} respectively denote the average results of classification tasks and regression tasks.

- **Uni-modal model**: For models that only accept SMILES input, we chose SMILES-Transformer as baselines. For models that only accept 2D graph information input, we chose PNA, CD-MVGNN, GROVER [51] and Mol-BERT as baselines.
- **Multi-modal model**: The following section will discuss the multi-modal model. GraphMVP conduct self-supervised learning between 2D topological structures and 3D geometric views. AdvProp uses 1D SMILES sequence and 2D graph information, respectively extracting features and combining them. 3DInfomax pretrains using both 2D graph information and 3D conformational information, enabling GNN to extract 3D information. MEMO combines molecular features from four perspectives.

4.5. Experiment results

The performance of our model and baseline for 8 datasets in MoleculeNet is summarized in Table 3. It can be seen that our model exhibits robust performance, achieving best results in six out of the eight datasets, with the average improvement rate 3.4% for classification tasks and 4.0% for regression tasks. This demonstrates the superiority of our model.

For a more comprehensive comparative analysis, we present the performance of our two branches (PremuNet-L and PremuNet-H). It is evident that our proposed PremuNet-L model also performs well, surpassing the previous best method on 4 datasets, which proves the effectiveness of our module, and further analysis on PremuNet-L is described in Section 5.1. By combining PremuNet-L with the high-dimensional information acquired from the PremuNet-H model, we achieve further improvements and obtain superior results on the majority of the datasets.

5. Ablation study

In order to clarify the contribution of each component in our model, we perform ablation analyses, which consist of two parts: the performance analysis of different molecular representations and the comparative analysis of different aggregation methods.

5.1. Molecular representations performance analysis of Premunet-L

In this section, we dissect the model modularly. We dissect the PremuNet-L branch into its components: Traditional-FP, Transformer-FP, and PNA+Atom-FP, which are molecular fingerprint extraction methods introduced in Sections 3.2.1, 3.2.2 and 3.2.4. They are evaluated on five classification datasets respectively. Fig. 3 shows the performance of these three components on five models. It can be observed that, overall, the performance of PremuNet-L is better than the three components without information fusion. This indicates that our three molecular representation methods have complementary informational characteristics, proving the necessity and effectiveness of multi-feature fusion mechanisms.

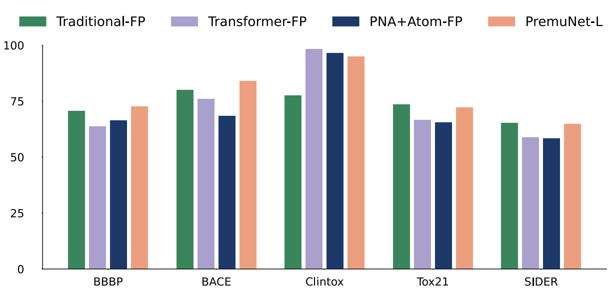

Fig. 3. Result of different molecular representations performance.

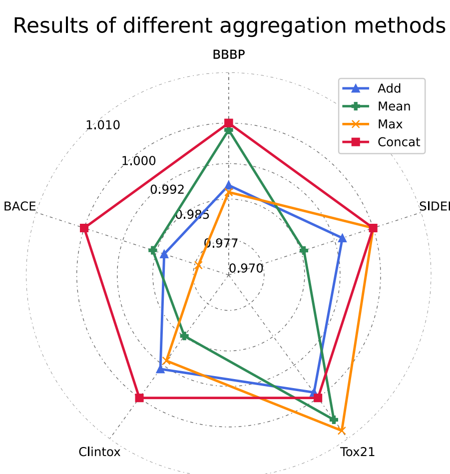

Fig. 4. Result of different aggregation methods. We take the result of the Concat method as a reference and calculate the ratio of the results of other methods to the result of Concat. The obtained results are shown in the figure.

5.2. Comparison of module aggregation methods

The performance of different aggregation methods of PremuNet-L and PremuNet-H, including Concat, Min, Max and Sum, is reported in Fig. 4. The results showed that, based on the Concat method, other aggregation methods perform worse in most cases, while Concat shows good generalization and achieves optimal results in most cases. This suggests that Concat can better integrate information of different modalities compared to Max, Mean and Sum. Other aggregation methods are occasionally better than Concat due to the bias of certain datasets and the different mixes of information fused by different aggregation methods, resulting in variations in performance. On balance, Concat is still the optimal choice.

6. Conclusion

In this paper, we introduce a new framework, PremuNet, for molecular representation learning. Our framework extracts both low-dimensional and high-dimensional features of molecules through two distinct branches, each employing a different encoder. Furthermore, we incorporate pre-training strategies and information fusion techniques specifically tailored for these encoders to enhance the predictive capabilities of downstream tasks. In the low-dimensional feature branch, we utilize three distinct encoders to extract low-dimensional features from three unique perspectives. Comprehensive performance evaluations on a publicly available molecular property prediction dataset demonstrate that our pre-trained multi-representation molecular framework effectively learns and combines features from various molecular representations, resulting in improvements over the current state-of-the-art on seven molecular prediction datasets (with an average improvement of 3.4% for classification tasks and 4.0% for regression tasks). This also substantiates the superiority of multi-modal molecular representation in predicting molecular properties.

Despite our success, there is still room for improvement. Our model possesses a relatively large number of parameters, necessitating significant computational resources, and the prediction speed warrants enhancement. Furthermore, there is no profound feature interaction between our low-dimensional and high-dimensional branches, only a rudimentary fusion. In future research, we may explore the correlation of features and incorporate parameter-sharing mechanisms to address these limitations.

CRediT authorship contribution statement

Haohui Zhang: Conceptualization, Software, Methodology, Writing – original draft. **Juntong Wu:** Software, Methodology, Formal analysis, Writing – original draft. **Shichao Liu:** Conceptualization, Supervision, Project administration, Writing – review & editing. **Shen Han:** Formal analysis, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Dictionary for converting token list to integer sequence

Key	Value	Key	Value	Key	Value
BEG	0	Ag	46	–	92
END	1	Au	47	@	93
PAD	2	Cd	48	.	94
UNK	3	Te	49	(	95
MSK	4	Pt	50	*	96
Sc	5	Sn	51	)	97
Na	6	Tb	52	$	98
Cr	7	c	53	:	99
s	8	Ir	54	0	100
o	9	Cl	55	1	101
Co	10	Ru	56	2	102
Mg	11	Re	57	3	103
Sm	12	Li	58	4	104
Si	13	Ga	59	5	105
te	14	Pd	60	6	106
K	15	Ge	61	7	107
V	16	b	62	8	108
C	17	Ba	63	9	109
Ho	18	Yb	64	@@	110
Mn	19	Ni	65	+1	111
N	20	S	66	−1	112
W	21	Sr	67	+2	113
Pb	22	p	68	−2	114
Se	23	Bi	69	+3	115
Ti	24	Rh	70	−3	116
Ac	25	Hg	71	+4	117
Mo	26	O	72	−4	118
P	27	Eu	73	+5	119
In	28	As	74	−5	120
Dy	29	Ca	75	+6	121
La	30	Zn	76	−6	122
H	31	Tl	77	+7	123
Y	32	se	78	−7	124
Tc	33	Cf	79	+8	125
Gd	34	n	80	−8	126
B	35	Fe	81	%10	127
Zr	36	Nd	82	%11	128
I	37	Sb	83	%12	129
Be	38	/	84	%13	130
Al	39	=	85	%14	131
Ra	40	+	86	%15	132
Cu	41	%	87	%16	133
Br	42	#	88	%17	134
F	43	[	89	%18	135
Cs	44	]	90	%19	136
U	45	\n	91		

Appendix B. Data and code availability

The datasets and splits of MoleculeNet in our experiment is available at http://snap.stanford.edu/ogb/data/graphproppred/csv_mol_download/. For the GEOM dataset, it can be obtained from https://github.com/learningmatter-mit/geom.

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